

Midterm 1 Question Bank

Lectures 1–4

83413 Introduction to VLSI

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Unless stated otherwise: NMOS device, long-channel square law, T = 300 K, $U_T \approx 26$ mV.

Coverage

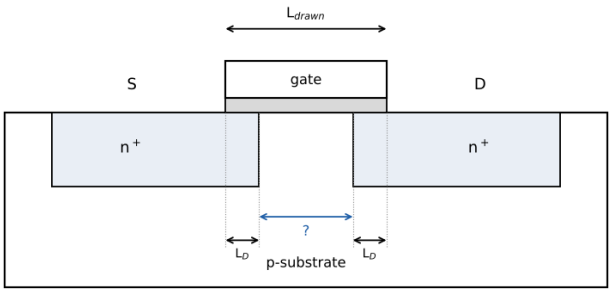
Lecture	Easy	Medium	Hard	Total
L1 Prerequisites Recap	6	13	2	21
L2 SPICE Modeling	2	2	0	4
L3 Transistor Impairments	5	15	4	24
L4 Performance Measures of a MOS	2	8	4	14
Total	15	38	10	63

Easy (Questions 1–15)

Question 1

L1 Prerequisites Recap · MOS structure · Easy

The source/drain junctions of a MOSFET side-diffuse under the gate by L_D on each side during fabrication. The effective channel length is:



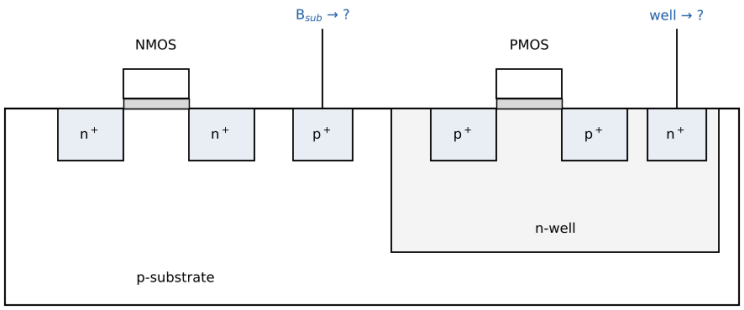
A. $L_{drawn} - 2L_D$
B. $L_{drawn} + 2L_D$
C. $L_{drawn} - L_D$
D. $L_{drawn} + L_D$

Answer: A. Each junction extends L_D under the gate from both ends, so $L_{eff} = L_{drawn} - 2L_D$.

Question 2

L1 Prerequisites Recap · Bulk terminal · Easy

In a CMOS circuit, to keep every source/drain junction diode reverse-biased, the NMOS p-substrate and the PMOS n-well are normally connected to:



A. Substrate to the most positive supply (V_{DD}), n-well to the most negative supply (ground)
B. Both to ground, the lowest potential in the circuit
C. Both to V_{DD}, the highest potential in the circuit
D. Substrate to the most negative supply (ground), n-well to the most positive supply (V_{DD})
E. Both left floating, so that the junctions carry no current

Answer: D. A p-n junction is reverse-biased when the p side is at the lower potential, so the p-substrate sits at the lowest potential in the system and the n-well at the highest. The substrate potential strongly affects device behaviour (body effect).

Question 3

L1 Prerequisites Recap · Small-signal parameters · Easy

For a MOSFET in saturation (square law, channel-length modulation neglected), which expression is NOT a valid form of g_m?

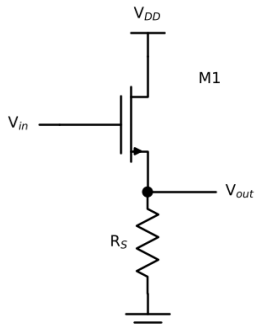
A. $\mu_n C_{ox}(W/L)\cdot V_{OV}$
B. $\sqrt{(2\mu_n C_{ox}(W/L)\cdot I_D)}$
C. I_D/V_{OV}
D. $2I_D/V_{OV}$

Answer: C. $g_m = \partial I_D/\partial V_{GS} = \mu_n C_{ox}(W/L)V_{OV}$; substituting $I_D = \tfrac{1}{2}\mu_n C_{ox}(W/L)V_{OV}^2$ gives $2I_D/V_{OV}$ and $\sqrt{(2\mu_n C_{ox}(W/L)I_D)}$. I_D/V_{OV} is half the true value.

Question 4

L1 Prerequisites Recap · Source follower · Easy

Which statement about the source follower (common-drain stage) with source resistor R_S is correct? Neglect channel-length modulation ($r_o \to \infty$) and the body effect.



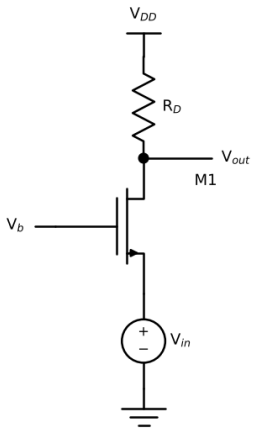
A. The gain is $-g_m R_S$, the output is inverted and $R_{out} \approx R_S$
B. The gain is $g_m R_S/(1 + g_m R_S) < 1$ and $R_{out} \approx (1/g_m) \parallel R_S$
C. The gain is slightly above 1 and $R_{out} \approx r_o \parallel R_S$
D. The input resistance is about $1/g_m$ and the gain is $+g_m R_S$

Answer: B. KVL and KCL on the small-signal model give $v_{out}/v_{in} = g_m R_S/(1 + g_m R_S)$; driving the output with a test source gives $R_{out} = (1/g_m) \parallel R_S$ (with channel-length modulation r_o appears in parallel as well). It is a voltage buffer: high input resistance, low output resistance, output in phase with the input.

Question 5

L1 Prerequisites Recap · Common-gate stage · Easy

For a common-gate stage with load R_D (gate at AC ground, body effect neglected, $r_o \gg R_D$), the input resistance and the sign of the voltage gain are approximately:



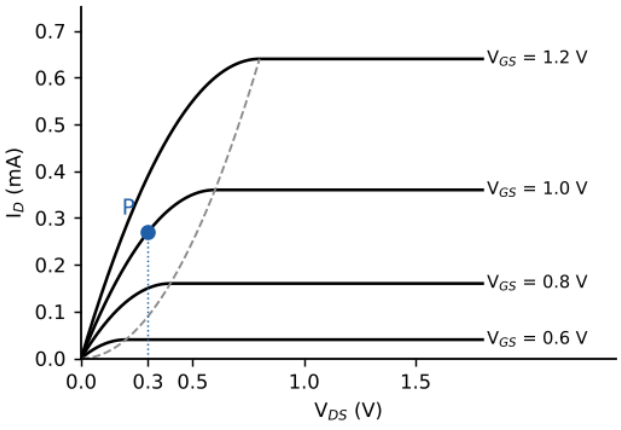
A. $R_{in} \to \infty$, gain $-g_m R_D$
B. $R_{in} \approx 1/g_m$, gain $+g_m R_D$
C. $R_{in} \approx R_D$, gain $+g_m R_D$
D. $R_{in} \approx 1/g_m$, gain $-g_m R_D$

Answer: B. The input enters at the source, so $v_{gs} = -v_{in}$ and the drain current $g_m v_{in}$ flows through R_D: the output is in phase with the input, $A_v = +g_m R_D$. The low input resistance ($\approx 1/g_m$) makes the stage naturally current-driven.

Question 6

L1 Prerequisites Recap · I–V curves · Easy

The figure shows the output characteristics of an NMOS with $V_{TH} = 0.4$ V (square law, $\lambda = 0$); the dashed curve is the saturation edge. At the marked point P ($V_{GS} = 1.0$ V, $V_{DS} = 0.3$ V) the transistor operates in:



A. Cut-off, since V_{DS} is below V_{TH}
B. Saturation, since $V_{DS} \geq V_{GS} - V_{TH}$
C. Triode, since $V_{DS} < V_{GS} - V_{TH}$
D. Subthreshold, since I_D is below its maximum

Answer: C. $V_{OV} = 0.6$ V is larger than $V_{DS} = 0.3$ V, so the channel is not pinched off: P lies to the left of the saturation edge ($I_D = \tfrac{1}{2}K \cdot V_{DS}^2$), on the rising part of the curve where I_D still depends strongly on V_{DS}.

Question 7

L2 SPICE Modeling · DC operating point · Easy

During a SPICE DC operating-point analysis, how are capacitors and inductors treated?

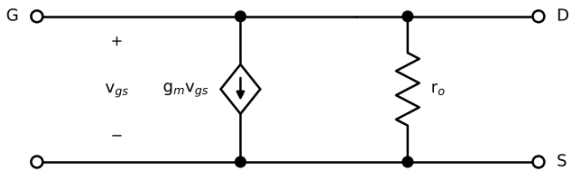
A. Capacitors as open circuits, inductors as short circuits
B. Capacitors as short circuits, inductors as open circuits
C. Both as open circuits
D. Both by their impedance at 1 Hz

Answer: A. At DC no current flows through a capacitor and no voltage drops across an inductor. The operating point must converge before any other analysis, and its node voltages seed every sweep.

Question 8

L2 SPICE Modeling · AC analysis · Easy

Which statement about SPICE AC analysis is correct?



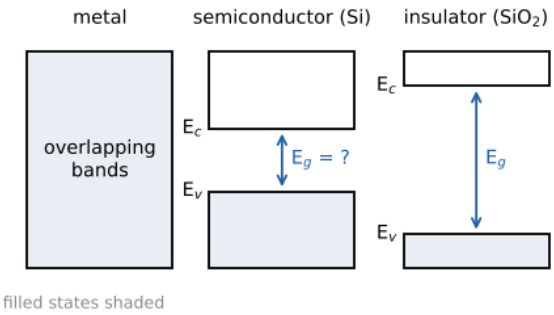
A. It captures the harmonic distortion of the transistors, because the full nonlinear models are evaluated at each frequency
B. It does not require a DC operating point, since the capacitors and inductors set the response
C. It integrates the circuit equations in time with the trapezoidal rule and then takes a Fourier transform
D. It is a small-signal linear analysis around the DC operating point, so no nonlinear effects appear

Answer: D. AC analysis first computes the DC operating point, replaces transistors and diodes by their small-signal models (g_m, r_o, capacitances) and solves a complex linear system (magnitude and phase) at each frequency.

Question 9

L3 Transistor Impairments · Semiconductor properties · Easy

The band-gap energy of silicon near room temperature is approximately:



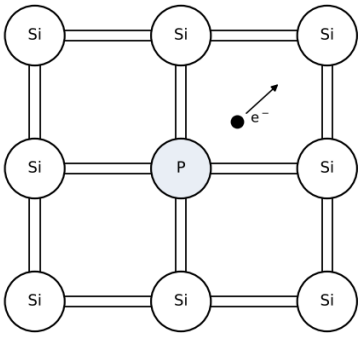
A. 0.66 eV
B. 1.12 eV
C. 5.47 eV
D. 9 eV

Answer: B. 0.66 eV is germanium, 5.47 eV diamond and about 9 eV SiO_2. Silicon's 1.12 eV is about 43kT at 300 K (42kT for the rounded 1.1 eV used in the slide), which leaves only about 1 in 10^{12} electrons in the conduction band.

Question 10

L3 Transistor Impairments · Doping · Easy

Doping silicon with phosphorus (group V, five valence electrons) produces:



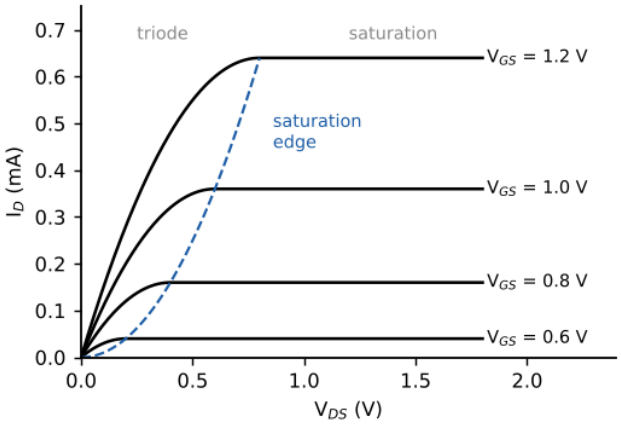
A. p-type material, each P atom contributing a hole and leaving a fixed negative ion
B. Intrinsic material, since each P atom creates one electron–hole pair
C. n-type material, each P atom contributing a conduction electron and leaving a fixed positive ion
D. An insulator, since the extra electrons fill the conduction band

Answer: C. The fifth electron is weakly bound and is ionized at room temperature: an electron–ion pair, not an electron–hole pair. Boron (group III) accepts an electron and gives p-type material.

Question 11

L3 Transistor Impairments · I–V derivation · Easy

In the long-channel model, the minimum V_{DS} for saturation ($V_{D,sat}$) is:



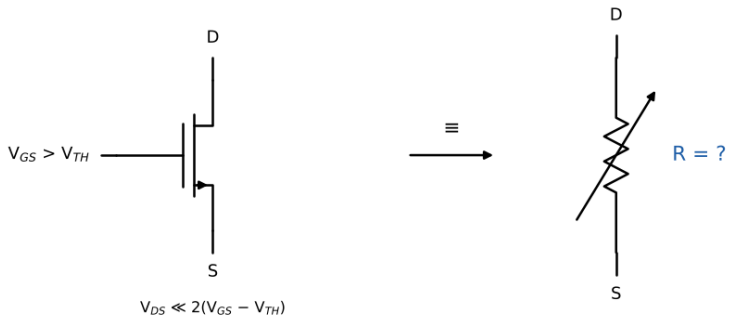
A. $V_{GS} - V_{TH}$
B. V_{TH}
C. V_{GS}
D. $2(V_{GS} - V_{TH})$
E. $V_{DD}/2$

Answer: A. An inversion layer exists only where $V_{GS} - V(x) > V_{TH}$. When V_{DS} reaches $V_{GS} - V_{TH}$ the channel pinches off at the drain end and I_D stops growing with V_{DS}.

Question 12

L3 Transistor Impairments · Deep triode · Easy

An NMOS driven into deep triode ($V_{DS} \ll 2(V_{GS} - V_{TH})$) behaves as a resistor R between drain and source, as in the figure. R equals:



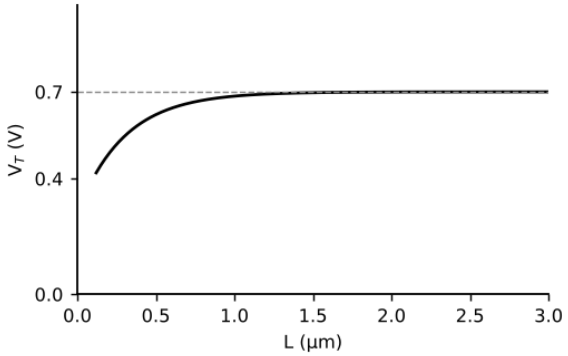
A. $\mu_n C_{ox}(W/L)(V_{GS} - V_{TH})$
B. $(V_{GS} - V_{TH})/I_D$
C. $1/(\lambda I_D)$
D. $1 / [\mu_n C_{ox}(W/L)(V_{GS} - V_{TH})]$

Answer: D. Dropping the V_{DS}^2 term leaves $I_D \approx \mu_n C_{ox}(W/L)(V_{GS} - V_{TH})V_{DS}$, linear in V_{DS}. With the lecture's values $\mu_n C_{ox}$ = 50 µA/V² and W/L = 10, an overdrive of 1 V gives R_{on} = 2 kΩ. The device is on even when it carries no current.

Question 13

L3 Transistor Impairments · Short-channel effects · Easy

The short-channel threshold roll-off that the lecture presents under drain-induced barrier lowering (DIBL) refers to:



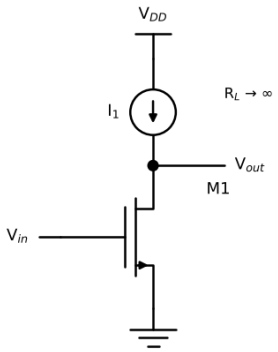
A. A decrease of V_T at short L, caused by the source/drain depletion regions protruding into the channel
B. An increase of V_T at short L, caused by the gate losing control of the channel
C. A decrease of mobility at high vertical field, caused by surface scattering
D. Saturation of the carrier velocity at high lateral field, caused by phonon scattering

Answer: A. The junction depletion regions take over part of the channel's depletion charge, so less gate voltage is needed to invert the surface. In the lecture plot V_t stays at 0.7 V for long devices and rolls off below roughly 1 µm.

Question 14

L4 Performance Measures of a MOS · Intrinsic gain · Easy

With channel-length modulation the transistor has a finite output resistance r_o. Its intrinsic gain, the maximum voltage gain of a common-source stage ($R_L \to \infty$), is:



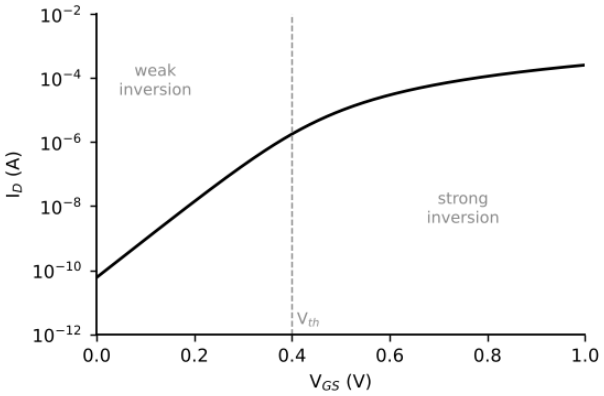
A. $g_m \cdot R_L$
B. $g_m \cdot r_o$
C. g_m / I_D
D. g_m / C_{gs}

Answer: B. $A_{DC,max} = g_m r_o = g_m / g_{ds}$. With $r_o = 1/(\lambda I_D)$ and $g_m = 2I_D/V_{OV}$, $g_m r_o \approx 2/(\lambda V_{OV})$: a device property independent of the load, and the quantity that limits the gain of op-amps built from the device.

Question 15

L4 Performance Measures of a MOS · Subthreshold swing · Easy

The subthreshold swing (SS) of a MOSFET is:



A. The change in I_D per volt of V_{GS} in strong inversion, in A/V
B. The slope of I_D versus V_{DS} in deep triode, i.e. $1/R_{on}$
C. The threshold-voltage shift per °C, about −2 mV/°C
D. The change in V_{GS} needed to change I_D by one decade in weak inversion, in mV/decade

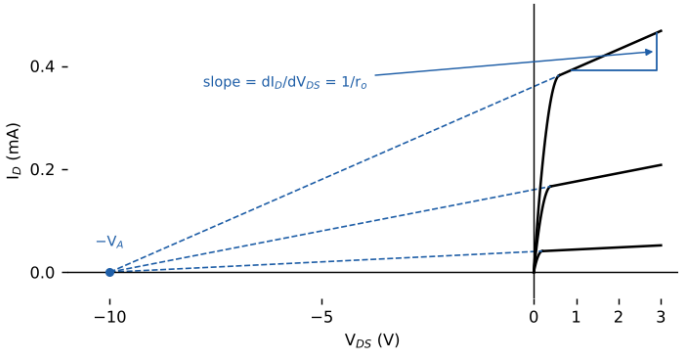
Answer: D. SS = $n \cdot U_T \cdot \ln 10 \approx 60 \cdot n$ mV/dec at 300 K, read from the $\log(I_D)$–V_{GS} curve below V_{th}. It sets the gate swing between the off and on states.

Medium (Questions 16–53)

Question 16

L1 Prerequisites Recap · Channel-length modulation · Medium

With channel-length modulation, $I_D = I_{D0}(1 + \lambda V_{DS})$. Which statement is correct?



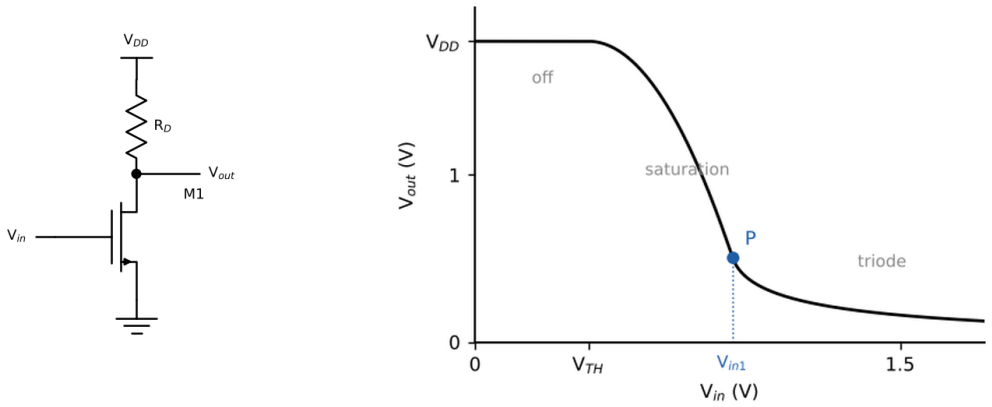
A. $r_o \approx \lambda I_D$ and $V_A = \lambda$, so r_o rises with bias current
B. The slope dI_D/dV_{DS} is the same for every bias current, so r_o is bias-independent
C. $r_o \approx 1/(\lambda I_D)$ and the extrapolated saturation curves meet at $V_{DS} = -V_A = -1/\lambda$
D. The common intercept $-V_A$ depends on V_{GS}, so it cannot serve as a device parameter
E. λ increases with L, so long devices have lower gain

Answer: C. $r_o = \partial V_{DS}/\partial I_D = 1/(\lambda I_{D0}) \approx 1/(\lambda I_D)$. Each curve's slope λI_{D0} grows with bias, while the intercept $-1/\lambda$ is a device constant (the 'pencil' of lines about $-V_A$). $\lambda \propto 1/L$, so longer transistors give higher r_o and gain.

Question 17

L1 Prerequisites Recap · Common-source stage · Medium

In the common-source stage with resistive load R_D shown in the figure, M1 leaves saturation and enters triode when V_{in} exceeds V_{in1} (point P on the transfer curve). At that boundary:



A. $V_{out} = V_{in1}$

B. $V_{out} = V_{in1} - V_{TH}$

C. $V_{out} = V_{DD}$

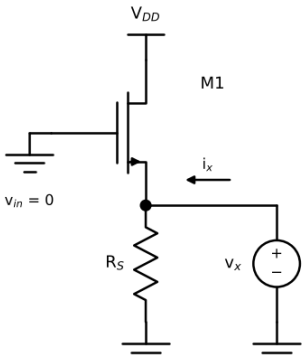
D. $V_{out} = V_{TH}$

Answer: B. Here $V_{out} = V_{DS}$ and $V_{in} = V_{GS}$; the saturation edge is $V_{DS} = V_{GS} - V_{TH}$, so the boundary point on the transfer curve sits at $V_{out} = V_{in1} - V_{TH}$. Beyond it the stage becomes a strongly nonlinear resistive divider.

Question 18

L1 Prerequisites Recap · Source follower · Medium

A source follower has $g_m = 5$ mS, $r_o = 10$ kΩ and $R_S = 2$ kΩ (body effect neglected). Its exact small-signal output resistance is:



A. 179 Ω

B. 200 Ω

C. 1.67 kΩ

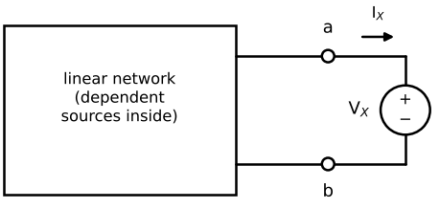
D. 12.2 kΩ

Answer: A. $R_{out} = (1/g_m) \parallel r_o \parallel R_S = 200\ \Omega \parallel 10\ k\Omega \parallel 2\ k\Omega = 1/(5 + 0.1 + 0.5)$ mS ≈ 179 Ω. 200 Ω is the $1/g_m$ approximation, valid when both r_o and R_S are much larger than $1/g_m$.

Question 19

L1 Prerequisites Recap · Thévenin and Norton · Medium

A one-port contains resistors, independent sources and a dependent (controlled) source. The correct way to find its Thévenin resistance R_{TH} is:



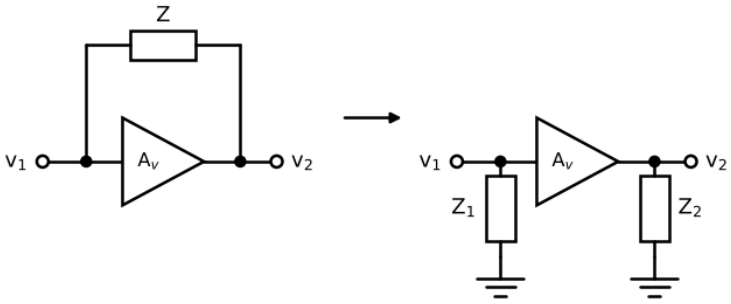
A. Zero every source, dependent ones included, and compute the resistance seen at the port
B. Replace the dependent source by an open circuit and zero the independent sources
C. Zero only the independent sources, keep the dependent source active, and drive the port with a test source: $R_{TH} = V_X/I_X$
D. R_{TH} is undefined when dependent sources are present; only the Norton form can be used

Answer: C. Dependent sources are part of the linear network's response and stay active; independent voltage sources become shorts and current sources become opens. V_{TH} is the open-circuit voltage, $I_N = V_{TH}/R_{TH}$ and $R_N = R_{TH}$.

Question 20

L1 Prerequisites Recap · Miller's theorem · Medium

Miller's theorem replaces a floating impedance Z between two nodes with voltage gain $A_v = v_o/v_i$ by two grounded impedances. Which statement is correct?



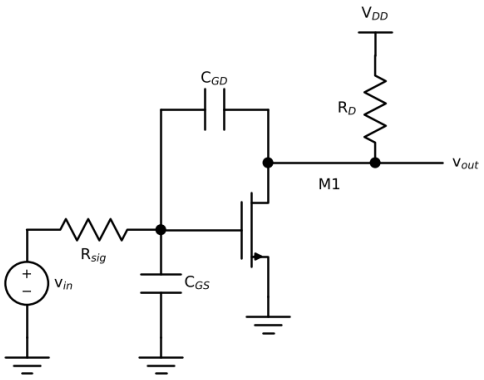
A. $Z_1 = Z(1 - A_v)$ at the input and $Z_2 = Z(1 - 1/A_v)$ at the output, with A_v the gain measured with Z removed
B. $Z_1 = Z{\cdot}A_v$ at the input and $Z_2 = Z/A_v$ at the output
C. The equivalence also preserves the internal currents of the circuit
D. $Z_1 = Z/(1 - A_v)$ at the input and $Z_2 = Z/(1 - 1/A_v)$ at the output, with A_v the gain measured with Z in place
E. It applies even when Z is the only signal path between the two nodes

Answer: D. The current drawn from node 1 is $(v_1 - v_2)/Z = v_1(1 - A_v)/Z$, hence $Z_1 = Z/(1 - A_v)$; the same argument at node 2 gives Z_2. Only the two node currents are reproduced, and the theorem fails when removing Z changes A_v.

Question 21

L1 Prerequisites Recap · Miller's theorem · Medium

A common-source stage has inverting gain $|A| = 10$, $C_{GD} = 20$ fF and $C_{GS} = 50$ fF. By Miller's theorem the total input capacitance is:



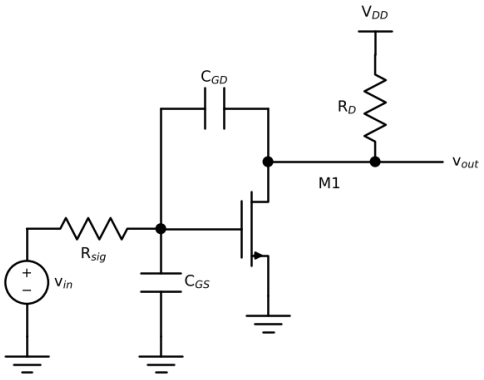
A. 70 fF
B. 90 fF
C. 220 fF
D. 250 fF
E. 270 fF

Answer: E. $C_1 = (1 + A)C_{GD} = 11 \times 20$ fF $= 220$ fF, so $C_{in} = C_{GS} + C_1 = 270$ fF, 3.9× the 70 fF obtained by ignoring Miller. On the output side $C_2 = (1 + 1/A)C_{GD} = 22$ fF $\approx C_{GD}$.

Question 22

L1 Prerequisites Recap · Miller's theorem · Medium

A common-source stage is driven from $R_{sig} = 2$ kΩ and has inverting gain $|A| = 15$, $C_{GS} = 40$ fF and $C_{GD} = 10$ fF. The input pole frequency $f_p = 1/(2\pi R_{sig} C_{in})$ is approximately:



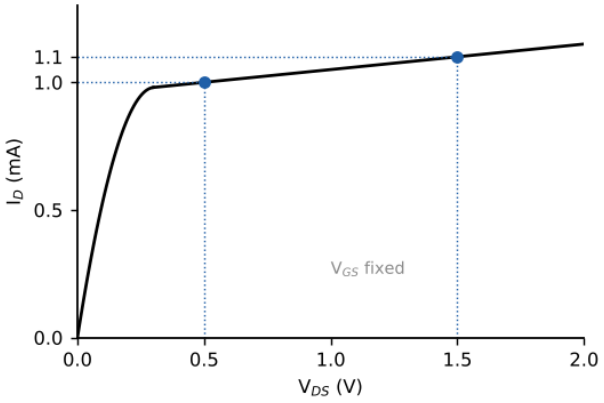
A. 400 MHz
B. 800 MHz
C. 1.6 GHz
D. 4 GHz

Answer: A. $C_{in} = C_{GS} + (1 + A)C_{GD} = 40 + 160 = 200$ fF, so $f_p = 1/(2\pi \cdot 2\ \text{k}\Omega \cdot 200\ \text{fF}) \approx 398$ MHz. Ignoring the Miller multiplication ($C_{in} = 50$ fF) would predict 1.6 GHz, four times too optimistic.

Question 23

L1 Prerequisites Recap · I–V curves · Medium

The figure shows one output curve of a MOSFET at fixed V_{GS}. In saturation it passes through (0.5 V, 1.00 mA) and (1.5 V, 1.10 mA). The small-signal output resistance r_o and the Early voltage V_A are approximately:



A. $r_o \approx 1\ k\Omega$, $V_A \approx 1$ V

B. $r_o \approx 100\ k\Omega$, $V_A \approx 95$ V

C. $r_o \approx 10\ k\Omega$, $V_A \approx 9.5$ V

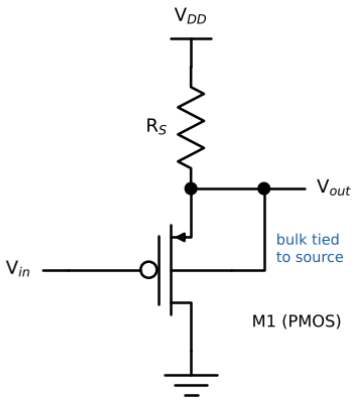
D. $r_o \approx 10\ k\Omega$, $V_A \approx 0.1$ V

Answer: C. $r_o = \Delta V_{DS}/\Delta I_D = 1\text{ V} / 0.1\text{ mA} = 10\ k\Omega$. Extrapolating the saturation line to $I_D = 0$ gives $V_A = I_D \cdot r_o - V_{DS} = 1.0\text{ mA} \times 10\ k\Omega - 0.5\text{ V} = 9.5$ V, i.e. $\lambda = 1/V_A \approx 0.1\ V^{-1}$.

Question 24

L1 Prerequisites Recap · Source follower (PMOS) · Medium

The source follower is rebuilt with a PMOS transistor whose bulk is tied to its source (figure), with R_S returned to V_{DD} and the drain at ground. Compared with the NMOS follower, which statement is correct?



A. The output is level-shifted downward, $V_{out} = V_{in} - |V_{GS}|$, exactly as in the NMOS follower, since only the carrier type changes

B. The output is level-shifted upward, $V_{out} = V_{in} + |V_{GS}|$, and because the PMOS sits in its own n-well the body can be tied to the source, so $V_{SB} = 0$ and there is no body effect

C. The body effect is stronger, because the PMOS source now moves with respect to the n-well potential

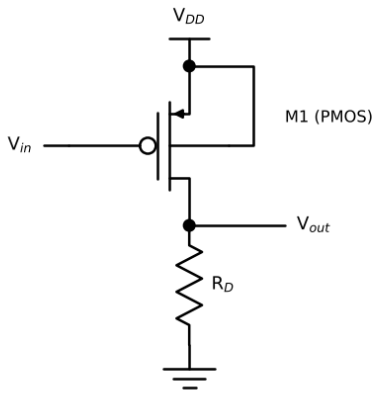
D. The gain exceeds 1, because the lower hole mobility raises $1/g_m$

Answer: B. An NMOS in the common p-substrate has its bulk at the most negative supply, so a follower whose source moves suffers body effect (V_{TH} rises with V_{SB}). A PMOS is fabricated in an n-well that can be connected to its own source instead of V_{DD}, which removes the body effect, and its source sits $|V_{GS}|$ above the gate, which suits an input biased near ground. The gain stays $g_mR_S/(1 + g_mR_S) < 1$; the lower hole mobility only means a wider device for the same g_m.

Question 25

L1 Prerequisites Recap · Common source (PMOS) · Medium

The common-source stage is rebuilt with a PMOS transistor of the same W/L (figure): the source sits at V_{DD}, R_D returns the drain to ground, and the device is biased at the same $|V_{OV}|$ as the NMOS version. With $\mu_p \approx 0.5 \cdot \mu_n$, compared with the NMOS stage:



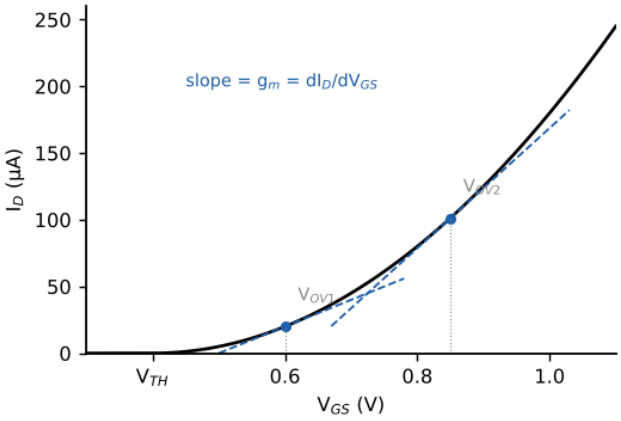
A. $|I_D|$ drops to half but g_m is unchanged, since g_m depends only on V_{OV}
B. g_m drops to half but $|I_D|$ is unchanged, so the gain halves at the same power
C. Nothing changes except the sign of the gain, which becomes positive because the PMOS pulls the output up
D. $|I_D|$ and g_m both drop to about half, so $|A_v| = g_m R_D$ halves; restoring g_m takes roughly twice the width, which doubles C_{GS}

Answer: D. $I_D = \frac{1}{2}\mu C_{ox}(W/L)V_{OV}^2$ and $g_m = \mu C_{ox}(W/L)V_{OV}$ are both proportional to mobility, so at equal geometry and overdrive both halve. The stage is still inverting: a rising gate reduces $|V_{GS}|$, the current through R_D falls and V_{out} across R_D falls. Doubling W restores g_m at the cost of $C_{GS} \propto W$, so the PMOS stage is slower, which is why NMOS is preferred where possible.

Question 26

L1 Prerequisites Recap · Small-signal parameters · Medium

How does the transconductance of a MOSFET in saturation (square law) change with the overdrive $V_{OV} = V_{GS} - V_{TH}$ (figure)?



A. g_m rises as V_{OV}^2, in proportion to I_D

B. g_m is independent of V_{OV}, because it is set by $\mu C_{ox} W/L$ alone

C. At fixed W/L, $g_m = \mu C_{ox}(W/L)V_{OV}$ rises linearly with V_{OV} (while I_D rises as V_{OV}^2); at fixed I_D, $g_m = 2I_D/V_{OV}$ falls as $1/V_{OV}$

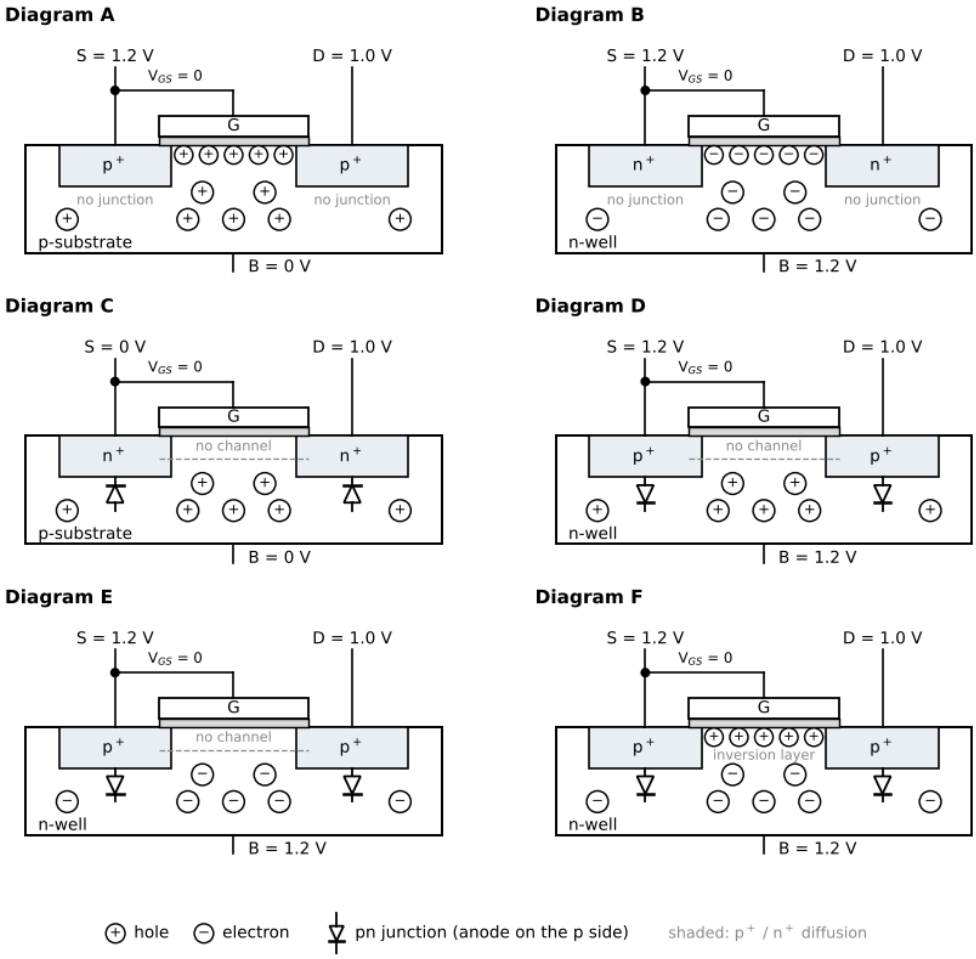
D. At fixed W/L, g_m falls as $1/V_{OV}$; at fixed I_D it rises linearly with V_{OV}

Answer: C. g_m is the slope of the I_D–V_{GS} parabola, which steepens linearly with overdrive: doubling V_{OV} doubles g_m but quadruples I_D, so $g_m/I_D = 2/V_{OV}$ falls. The same square law read at constant current gives $g_m = 2I_D/V_{OV}$: a wider device biased at lower overdrive has more transconductance per ampere. Velocity saturation eventually flattens g_m at $W \cdot C_{ox} \cdot v_{sat}$, and in weak inversion $g_m = I_D/(nU_T)$ no longer depends on V_{OV}.

Question 27

L1 Prerequisites Recap · MOS structure (PMOS) · Medium

The six diagrams show candidate cross-sections, each with the source, drain and bulk voltages marked and the gate tied to the source ($V_{GS} = 0$). The mobile carriers are drawn as circled signs (+ holes, − electrons) and every pn junction as a diode. Which diagram represents a PMOS transistor with the correct charge-carrier state for the marked voltages?



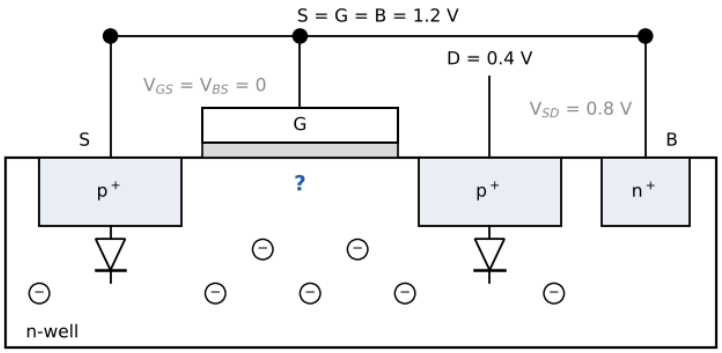
A. Diagram A
B. Diagram B
C. Diagram C
D. Diagram D
E. Diagram E
F. Diagram F

Answer: E. Diagram E: p^+ source and drain in an n-well whose majority carriers are electrons, the well at the highest potential so that the source–well junction is unbiased and the drain–well junction reverse-biased, and no hole channel at $V_{GS} = 0 > V_{TH}$: the device is off, since the well's electrons cannot pass between the p^+ terminals and the holes that could, the well's minority carriers, only appear as an inversion layer for $V_{GS} < V_{TH} < 0$. Diagrams A (p^+ in a p-substrate) and B (n^+ in an n-well) contain no junction: they are p- and n-type resistors, not transistors. Diagram C is an NMOS: holes in a p-substrate, electron channel when on. Diagram D draws holes as the mobile carriers of an n-well, which would make it p-type; in an n-well holes are minority carriers at n_i^2/N_D. Diagram F draws an inversion layer of holes at $V_{GS} = 0$, which an enhancement PMOS only forms for $V_{GS} < V_{TH}$ (a depletion-mode device would need a built-in p implant).

Question 28

L1 Prerequisites Recap · MOS structure (PMOS) · Medium

The PMOS in the figure (p^+ source and drain in an n-well, $V_{TH} = -0.4$ V) has its gate and bulk tied to the source, so $V_{GS} = V_{BS} = 0$, while the drain is held 0.8 V below the source ($V_{SD} = 0.8$ V). The well's majority carriers are drawn; the channel region is left open. Is the transistor conducting?



A. No: at $V_{GS} = 0 > V_{TH}$ there is no inversion layer, the source–well junction is unbiased and the drain–well junction is reverse-biased by 0.8 V, so only junction leakage and a subthreshold current several decades below the on-current flow

B. Yes: $V_{SD} = 0.8$ V exceeds $|V_{TH}|$, so the drain–source voltage forms the channel and the device conducts in saturation

C. Yes: the 0.8 V drop forward-biases the source–well junction and holes cross the well to the drain

D. Yes, weakly: $V_{GS} = 0$ lies only 0.4 V above V_{TH}, so a subthreshold current comparable to the on-current flows

Answer: A. A hole channel needs the gate at least $|V_{TH}| = 0.4$ V below the source ($V_{GS} < V_{TH}$); the drain voltage cannot create it. With the well tied to the source, the source–well diode sees zero bias and the drain–well diode (p^+ drain at 0.4 V, well at 1.2 V) is reverse-biased, so the well's electrons cannot reach the drain either. V_{GS} sits 0.4 V above threshold: with SS ≈ 80 mV/dec that is about five decades of subthreshold attenuation, i.e. an off-state leakage, not conduction. Tying the gate to the bulk and source is exactly how a PMOS is held off.

Question 29

L2 SPICE Modeling · Transient analysis · Medium

Which statement about SPICE transient analysis is correct?

A. It starts from zero initial conditions on every node and integrates forward with a fixed time step, without any Newton–Raphson iteration

B. It linearizes the transistors once at the operating point and keeps those models fixed for the whole run

C. It computes each time point independently of the previous one, so the time step has no effect on accuracy

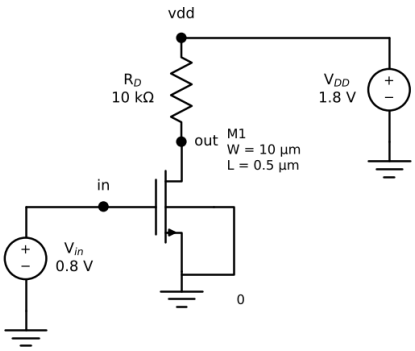
D. It starts from the DC operating point, replaces C and L at each time step by integration-derived equivalents, and solves each step with Newton–Raphson seeded by the previous point

Answer: D. The equivalent model of each reactive element changes at every time point. As in a DC sweep, seeding Newton–Raphson with the previous solution reduces the iteration count.

Question 30

L2 SPICE Modeling · Netlist · Medium

The circuit in the figure (nodes vdd, in, out and ground 0; M1 is a 1.2 V core NMOS with W = 10 µm, L = 0.5 µm) is to be simulated in Spectre, whose MOS pin order is drain, gate, source, bulk. Which netlist represents the circuit correctly?



A.
```
vdd (vdd 0) vsource dc=1.8
vin (in 0) vsource dc=0.8
rd (vdd out) resistor r=10k
m1 (in out 0 0) nmos1v w=10u l=0.5u
```

B.
```
vdd (vdd 0) vsource dc=1.8
vin (in 0) vsource dc=0.8
rd (vdd out) resistor r=10k
m1 (out in 0 0) nmos1v w=10u l=0.5u
```

C.
```
vdd (vdd 0) vsource dc=1.8
vin (in 0) vsource dc=0.8
rd (out 0) resistor r=10k
m1 (out in 0 0) nmos1v w=10u l=0.5u
```

D.
```
vdd (vdd 0) vsource dc=1.8
vin (in 0) vsource dc=0.8
rd (vdd out) resistor r=10k
m1 (out in 0 vdd) nmos1v w=10u l=0.5u
```

E.
```
vdd (vdd 0) vsource dc=1.8
vin (0 in) vsource dc=0.8
rd (vdd out) resistor r=10k
m1 (out in 0 0) nmos1v w=10u l=0.5u
```

Answer: B. A Spectre instance line reads name (d g s b) model parameters, as in the lecture's mn1 (d g 0 0) nch: the drain is node out, the gate is node in, and source and bulk sit at 0, with R_D between vdd and out. The wrong netlists swap gate and drain (m1 (in out 0 0)), return R_D to ground (rd (out 0)), tie the NMOS bulk to vdd (m1 (out in 0 vdd)), which forward-biases the body diode, or reverse the input source (vin (0 in)), which puts the gate at −0.8 V.

Question 31

L3 Transistor Impairments · Intrinsic carriers · Medium

The intrinsic carrier concentration of silicon is $n_i \approx 1 \times 10^{10}$ cm^{-3} at 300 K. Doubling the absolute temperature to 600 K increases n_i by roughly:

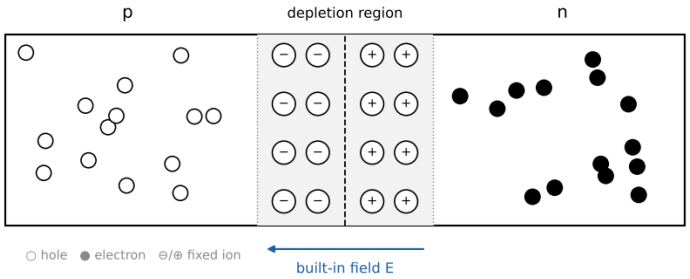
A. 2×
B. 8×
C. 10^3×
D. 10^5×

Answer: D. $n_i = 5.2 \times 10^{15} \cdot T^{3/2} \cdot \exp(-E_g/2kT)$: the exponential term dominates, giving about five orders of magnitude (to about 10^{15} cm^{-3}) for a factor of two in T, which is why semiconductor behaviour is so sensitive to heat.

Question 32

L3 Transistor Impairments · pn junction · Medium

In a pn junction at equilibrium, which statement is correct?



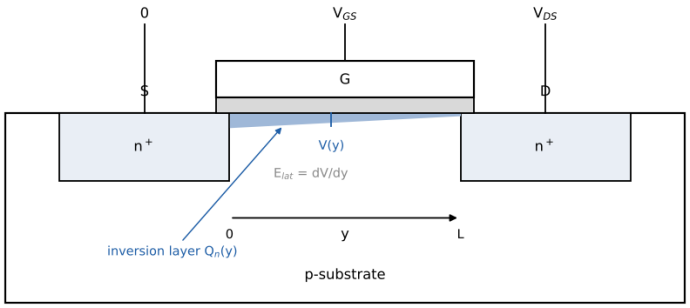
A. Mobile electrons and holes inside the depletion region carry a steady built-in current around the loop
B. The built-in field drives a net diffusion current from the n side to the p side
C. Fixed ionized dopants form a space charge whose field balances carrier diffusion; the Fermi level is flat across the junction
D. Forward bias widens the depletion region and reverse bias narrows it

Answer: C. Electrons diffusing from the n side recombine with holes, leaving fixed ions behind (negative on the p side, positive on the n side); the resulting field stops further net transfer and the bands bend across the depletion region. Reverse bias widens the region and only the leakage I_S flows until breakdown.

Question 33

L3 Transistor Impairments · I–V derivation · Medium

The first-order derivation of the MOS I–V characteristics (figure) rests on three assumptions. Which set states them correctly?



A. (1) The current is carried by the depletion charge; (2) the lateral field sets the channel charge; (3) the carrier velocity is constant at v_{sat} along the channel

B. (1) The current is carried by the mobile inversion charge; (2) the vertical field sets that charge through the local gate-to-channel voltage, the gradual-channel approximation; (3) the carrier velocity is proportional to the lateral field, v = μE

C. (1) The current is driven by diffusion; (2) the channel charge is set by the substrate doping N_{sub}; (3) the carrier velocity is proportional to the vertical field

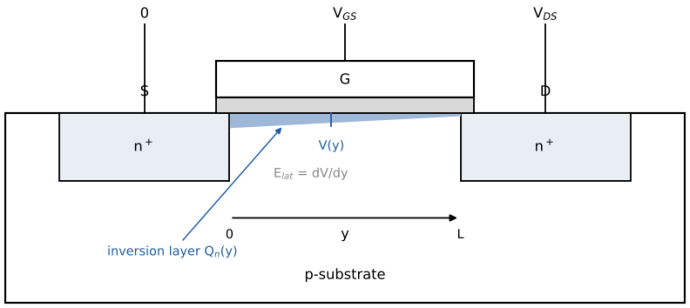
D. (1) The current is carried by the inversion charge; (2) the channel charge is uniform along the channel; (3) the carrier velocity saturates at the source end

Answer: B. Assumption 1 gives $I_D = Q_n \cdot v \cdot W$; assumption 2 says the vertical field dominates, so the local charge follows the local overdrive, $Q_n(y) = C_{ox}[V_{GS} - V(y) - V_t]$; assumption 3 is Ohm's law for the carriers and is the first to fail in short channels (velocity saturation). Integrating $I_D \cdot dy = W\mu C_{ox}[V_{GS} - V(y) - V_t] \cdot dV$ from 0 to L and from 0 to V_{DS} gives the triode equation.

Question 34

L3 Transistor Impairments · I–V derivation · Medium

Along the channel of an NMOS in triode (figure) the channel potential V(y) rises from 0 at the source to V_{DS} at the drain. Which statement about the inversion charge per unit area $Q_n(y)$ is correct?



A. $Q_n(y) = C_{ox}[V_{GS} - V(y) - V_t]$ is largest at the source and decreases toward the drain, where the gate-to-channel voltage is smallest; it reaches zero at the drain end when $V_{DS} = V_{GS} - V_t$ (pinch-off)
B. $Q_n(y) = C_{ox}[V_{GS} - V_t]$ everywhere, because the gate voltage is the same along the whole channel
C. $Q_n(y)$ is smallest at the source and grows toward the drain, where the lateral field is strongest
D. $Q_n(y) = C_{ox} \cdot V(y)$: the charge is proportional to the local channel potential, so it is zero at the source

Answer: A. The gate induces inversion relative to the local channel potential, $V_{GC}(y) = V_{GS} - V(y)$. Since V(y) rises toward the drain, the local overdrive and the induced charge fall; the same current must then be carried by fewer carriers, so the carrier velocity, and with it the lateral field dV/dy, rise toward the drain. When $V_{GS} - V(L)$ drops to V_t the channel pinches off at the drain and the current stops growing with V_{DS} (saturation).

Question 35

L3 Transistor Impairments · Threshold voltage · Medium

In $V_T = \Phi_{MS} + 2\Phi_F + Q_{dep}/C_{ox} - Q_{ox}/C_{ox}$, the positive fixed oxide/interface charge Q_{ox}:

A. Raises V_T by Q_{ox}/C_{ox}, since the extra charge must be imaged by the gate
B. Has no effect, because the gate screens charge located at the interface
C. Lowers V_T by Q_{ox}/C_{ox}, and its drift over time is one mechanism of transistor aging
D. Affects only PMOS devices, where the substrate is n-type

Answer: C. Positive charge at the Si–SiO_2 interface attracts electrons to the surface, so less gate voltage is needed for inversion: $\Delta V_T = -Q_{ox}/C_{ox} \propto -Q_{ox} \cdot t_{ox}$.

Question 36

L3 Transistor Impairments · Threshold voltage · Medium

Which change raises the magnitude of the threshold voltage $|V_T|$ of an NMOS transistor?

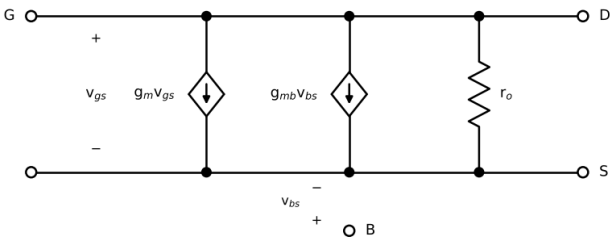
A. Lighter substrate doping and thinner gate oxide
B. Heavier substrate doping and thicker gate oxide
C. Higher operating temperature
D. More positive fixed oxide charge Q_{ox}

Answer: B. $\Phi_F \propto \ln(N_{sub}/n_i)$ and $Q_{dep} \propto \sqrt{N_{sub}}$ both grow with doping, and a thicker oxide lowers C_{ox} so Q_{dep}/C_{ox} grows. Temperature lowers $|V_T|$ by about 1–2 mV/°C and positive Q_{ox} lowers it as well.

Question 37

L3 Transistor Impairments · Body effect · Medium

The body transconductance g_{mb} of a MOSFET in saturation is:



A. Equal to g_m, since the bulk acts as a second gate

B. $g_m \cdot \gamma$, independent of V_{SB}

C. Zero whenever the device is in saturation

D. $g_m \cdot (2\Phi_F + V_{SB})$, growing with reverse body bias

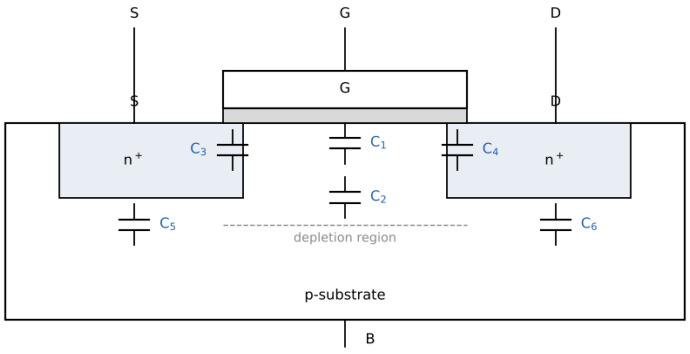
E. $\eta \cdot g_m$ with $\eta = \gamma / (2\sqrt{(2\Phi_F + V_{SB})})$, typically a fraction of g_m

Answer: E. $g_{mb} = \partial I_D/\partial V_{BS} = (\partial I_D/\partial V_{TH})(\partial V_{TH}/\partial V_{BS})$; with $V_{TH} = V_{TH0} + \gamma(\sqrt{(2\Phi_F + V_{SB})} - \sqrt{(2\Phi_F)})$ this gives $g_{mb} = g_m \cdot \gamma/(2\sqrt{(2\Phi_F + V_{SB})})$. The bulk acts as a second, weaker gate.

Question 38

L3 Transistor Impairments · Gate capacitance · Medium

The intrinsic (channel) gate capacitances of a MOSFET in saturation are:



A. $C_{GS} = \frac{2}{3}WLC_{ox}$, $C_{GD} \approx 0$ (overlap WC_{ov} only), $C_{GB} \approx 0$

B. $C_{GS} = \frac{1}{2}WLC_{ox}$, $C_{GD} = \frac{1}{2}WLC_{ox}$, $C_{GB} = 0$

C. $C_{GS} = WLC_{ox}$, $C_{GD} = WLC_{ox}$, $C_{GB} = 0$

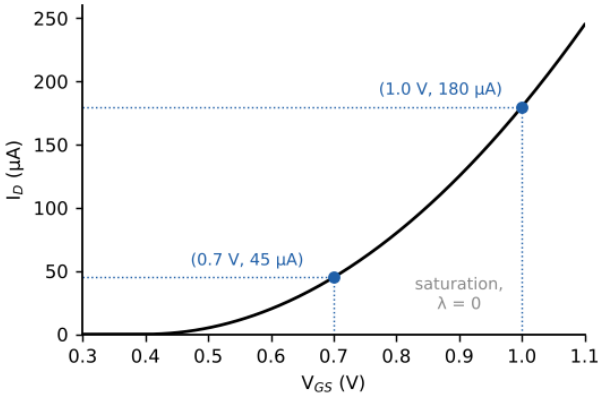
D. $C_{GS} = \frac{1}{3}WLC_{ox}$, $C_{GD} = \frac{1}{3}WLC_{ox}$, $C_{GB} = \frac{1}{3}WLC_{ox}$

Answer: A. In saturation the inversion charge tapers to about zero at the drain; integrating gives $Q_{ch} = \frac{2}{3}WLC_{ox}(V_{GS} - V_{TH})$, all supplied through the source. Beyond pinch-off the drain is isolated from the channel by a depleted region, so moving V_D only shifts the pinch-off point. The ½ / ½ split applies in triode.

Question 39

L3 Transistor Impairments · I–V curves · Medium

The figure shows I_D versus V_{GS} of an NMOS in saturation (square law, $\lambda = 0$): $I_D = 45\ \mu A$ at $V_{GS} = 0.7$ V and 180 µA at $V_{GS} = 1.0$ V. The threshold voltage and $K = \mu_n C_{ox}(W/L)$ are:



A. $V_{TH} = 0.4$ V, $K = 0.5$ mA/V²

B. $V_{TH} = 0.3$ V, $K = 1$ mA/V²

C. $V_{TH} = 0.55$ V, $K = 2$ mA/V²

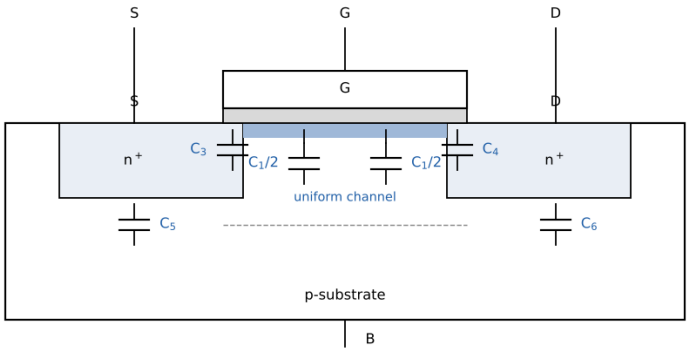
D. $V_{TH} = 0.4$ V, $K = 1$ mA/V²

Answer: D. $I_D \propto (V_{GS} - V_{TH})^2$: a 4× current ratio means a 2× overdrive ratio, $1.0 - V_{TH} = 2(0.7 - V_{TH})$, so $V_{TH} = 0.4$ V. Then $K = 2I_D/V_{OV}^2 = 2 \times 180\ \mu A/(0.6\ V)^2 = 1$ mA/V². (Equivalently, $\sqrt{I_D}$ versus V_{GS} is a straight line with intercept V_{TH}.)

Question 40

L3 Transistor Impairments · Gate capacitance · Medium

In the triode region ($V_{DS} \ll V_{GS} - V_{TH}$, nearly uniform channel, figure), the intrinsic gate capacitances of a MOSFET are:



A. $C_{GS} = \frac{2}{3}WLC_{ox}$, $C_{GD} \approx 0$, $C_{GB} \approx 0$

B. $C_{GS} = C_{GD} = 0$ (overlap only), $C_{GB} = WLC_{ox}$

C. $C_{GS} = WLC_{ox}$, $C_{GD} = 0$, $C_{GB} = 0$

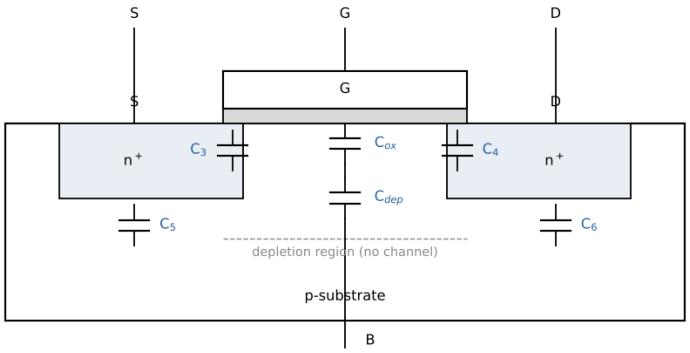
D. $C_{GS} = C_{GD} = \frac{1}{2}WLC_{ox}$, $C_{GB} \approx 0$

Answer: D. In triode the inversion layer is a nearly uniform conducting sheet connected to both source and drain, so the gate–channel capacitance WLC_{ox} (C_1) splits equally between the two terminals; the channel shields the bulk from the gate, so $C_{GB} \approx 0$. The overlap capacitances WC_{ov} add to both C_{GS} and C_{GD}.

Question 41

L3 Transistor Impairments · Gate capacitance · Medium

With the transistor off ($V_{GS} < V_{TH}$, no channel, figure), the gate capacitances of a MOSFET are:



A. C_{GS} and C_{GD} reduce to the overlap terms WC_{ov}, and C_{GB} is the series combination of the oxide capacitance WLC_{ox} and the depletion capacitance C_{dep}

B. $C_{GS} = C_{GD} = ½WLC_{ox}$ and $C_{GB} = 0$, as in triode

C. $C_{GS} = C_{GD} = C_{GB} = 0$, since no charge is stored without a channel

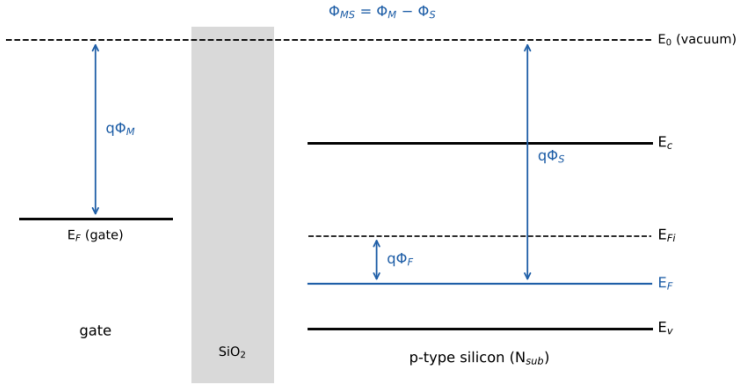
D. $C_{GB} = WLC_{ox} + C_{dep}$ (parallel combination) and $C_{GS} = C_{GD} = 0$

Answer: A. Without a channel the gate field terminates on the depletion charge in the substrate, so the gate sees C_{ox} in series with C_{dep} (a voltage divider between the oxide and the depletion region). A more negative V_{GS} shrinks the depletion region (accumulation) and C_{GB} rises toward WLC_{ox}; above V_{TH} the channel returns the full oxide capacitance, now to the source and drain.

Question 42

L3 Transistor Impairments · Threshold voltage · Medium

In the threshold-voltage expression $V_{TH} = \Phi_{MS} + 2\Phi_F + Q_{dep}/C_{ox}$ (figure), the terms Φ_{MS} and Φ_F are:



A. Φ_{MS} is the surface potential at the onset of inversion; Φ_F is the built-in potential of the source and drain junctions

B. Φ_{MS} is the work-function difference between the gate and the silicon substrate, fixed by the gate material; Φ_F is the bulk Fermi potential $(kT/q)\cdot\ln(N_{sub}/n_i)$, the separation between the Fermi level of the doped silicon and the intrinsic level

C. Φ_{MS} is the voltage dropped across the gate oxide at threshold; Φ_F is the flat-band voltage of the MOS capacitor

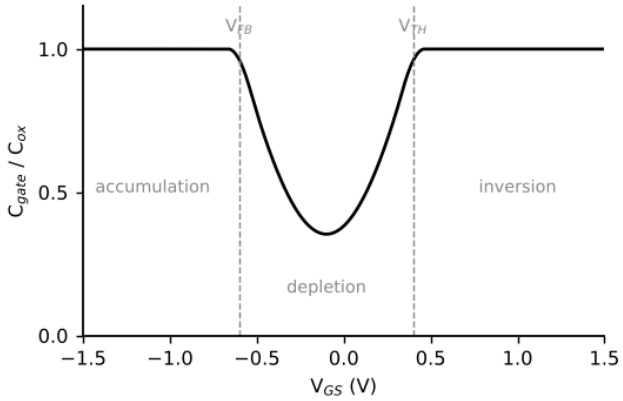
D. Φ_{MS} is the Schottky barrier of the gate contact; Φ_F is the thermal voltage kT/q

Answer: B. $\Phi_{MS} = \Phi_M - \Phi_S$ is set by the gate material (n^+/p^+ poly or metal) and is independent of N_{sub} and t_{ox}. Φ_F grows logarithmically with doping and with temperature through n_i; the surface must be bent by $2\Phi_F$ to become as strongly n-type as the bulk is p-type, which is why $2\Phi_F$ appears in V_{TH}.

Question 43

L3 Transistor Impairments · Gate capacitance · Medium

The figure shows the low-frequency gate capacitance per unit area of a MOS capacitor on a p-type substrate versus V_{GS}. Which statement is correct?



A. The capacitance is largest in depletion, because the depletion region adds its charge to the oxide charge

B. In strong inversion the capacitance equals C_{dep}, because the channel screens the oxide from the gate

C. In accumulation ($V_{GS} < V_{FB}$) and in strong inversion ($V_{GS} > V_{TH}$) the capacitance equals C_{ox}; in between it falls to the series combination of C_{ox} and the depletion capacitance, lowest where the depletion region is widest

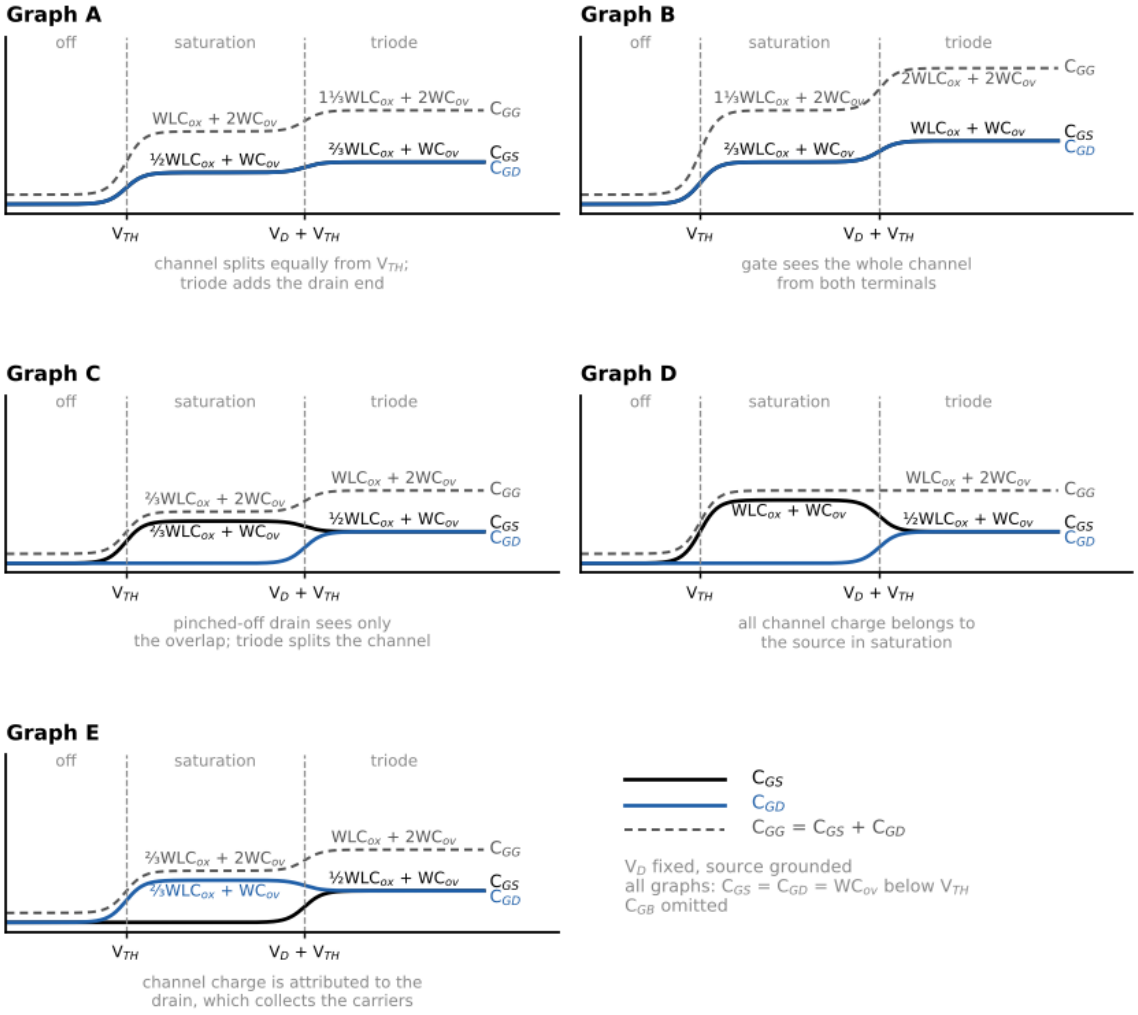
D. In accumulation the capacitance is close to zero, because the holes are repelled from the surface

Answer: C. A negative gate attracts holes to the interface (accumulation) and the gate sees the oxide alone, C_{ox}. Between flat band and threshold the surface is depleted, so C_{ox} is in series with a depletion capacitance that shrinks as the depletion region widens. Above V_{TH} the inversion layer supplies the charge and the capacitance returns to C_{ox}, now between gate and channel.

Question 44

L3 Transistor Impairments · Gate capacitance · Medium

The five annotated graphs show candidate curves of C_{GS} (black), C_{GD} (blue) and the total gate capacitance $C_{GG} = C_{GS} + C_{GD}$ (gray, dashed) versus V_{GS} for an NMOS at a fixed drain voltage V_D (source grounded), overlap capacitance WC_{ov} included and C_{GB} omitted. Each graph carries its capacitance levels and a one-line justification. Which graph is correct?



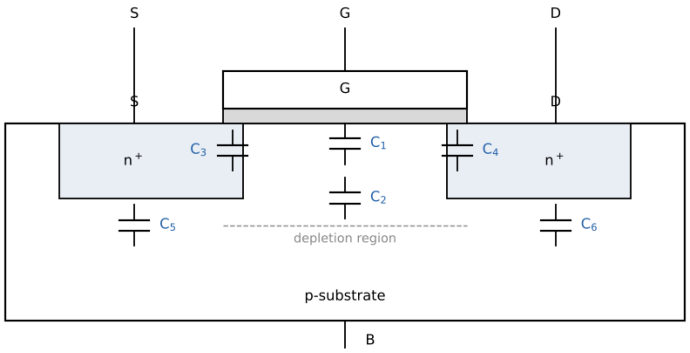
A. Graph A
B. Graph B
C. Graph C
D. Graph D
E. Graph E

Answer: C. Below V_{TH} only the overlaps remain. In saturation ($V_{TH} < V_{GS} < V_D + V_{TH}$) the channel is pinched off at the drain, so its charge is supplied by the source: $C_{GS} = \frac{2}{3}WL C_{ox} + WC_{ov}$ while C_{GD} stays at WC_{ov}. Once V_{GS} exceeds $V_D + V_{TH}$ ($V_{GD} > V_{TH}$) the drain end inverts, the device is in triode and the channel capacitance splits equally: $\frac{1}{2}WLC_{ox} + WC_{ov}$ each. C_{GG} therefore rises to $\frac{2}{3}WLC_{ox} + 2WC_{ov}$ in saturation and to $WLC_{ox} + 2WC_{ov}$ in triode, never beyond the oxide capacitance plus the overlaps. Graph A swaps the ½ and ⅔ between saturation and triode; Graph B lets the drain see the channel while it is pinched off, so C_{GG} exceeds $WLC_{ox} + 2WC_{ov}$; Graph D gives the source the whole WLC_{ox} instead of ⅔ of it; Graph E has the right C_{GG} but hands the $\frac{2}{3}WLC_{ox}$ to the pinched-off drain end, which holds no channel charge.

Question 45

L3 Transistor Impairments · Gate capacitance · Medium

The cross-section in the figure marks six capacitances C_1 to C_6 of an NMOS. Which assignment is correct?



A. C_1: gate–substrate capacitance; C_2: oxide capacitance; C_3, C_4: junction capacitances; C_5, C_6: gate overlap capacitances

B. C_1: gate–channel oxide capacitance; C_2: channel–substrate depletion capacitance; C_3, C_4: gate overlap of the source and drain; C_5, C_6: source–bulk and drain–bulk junction capacitances

C. C_1: gate overlap capacitance; C_2: gate–channel capacitance; C_3, C_4: depletion capacitances; C_5, C_6: gate–bulk capacitances

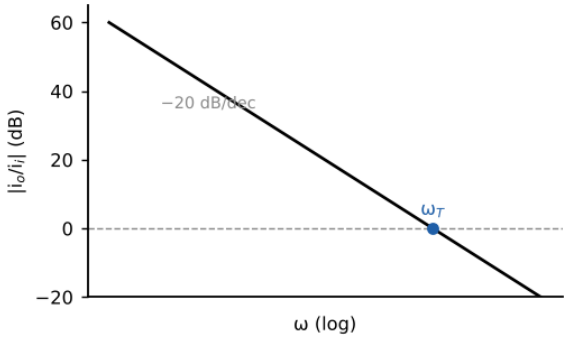
D. C_1: gate–channel capacitance; C_2: gate overlap capacitance; C_3, C_4: junction capacitances; C_5, C_6: channel–substrate capacitances

Answer: B. $C_1 = WLC_{ox}$ is the intrinsic gate–channel capacitance that is shared between C_{GS} and C_{GD} according to the region of operation; C_2 is the depletion capacitance between channel and substrate, which sets C_{GB} when no channel exists; C_3 and C_4 are the gate-poly overlaps of source and drain, WC_{ov} each and bias-independent to first order; C_5 and C_6 are the reverse-biased junction capacitances C_{SB} and C_{DB}, which depend on the source and drain voltages with respect to the substrate.

Question 46

L4 Performance Measures of a MOS · Transit frequency · Medium

For a long-channel MOSFET, $\omega_T = g_m/C_{gg} \approx (3/2)\cdot\mu V_{OV}/L^2$. Which statement is correct?



A. Raising V_{OV} raises ω_T at the cost of current; shrinking L raises ω_T at the cost of intrinsic gain

B. ω_T grows with L^2, so longer devices are faster

C. ω_T is independent of V_{OV}, since both g_m and C_{gg} scale with it

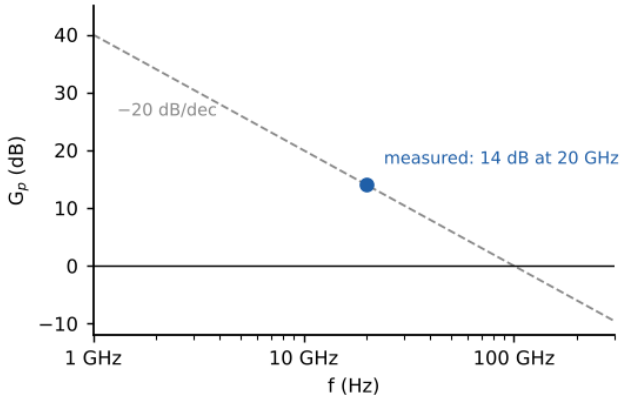
D. ω_T includes the effect of gate resistance, which is why it limits the power gain

Answer: A. ω_T is the frequency where the common-source current gain $|i_o/i_i| = g_m/(\omega C_{gg})$ falls to unity. It quantifies g_m/C_{gs} only; the lumped model is trusted up to about $\omega_T/5$, and the gate resistance enters ω_{max}, not ω_T.

Question 47

L4 Performance Measures of a MOS · Maximum oscillation frequency · Medium

The common-source power gain follows $G_p = (\omega_{max}/\omega)^2$, falling at 20 dB/decade. If a maximum power gain of 14 dB (25×) is measured at 20 GHz, f_{max} is approximately:



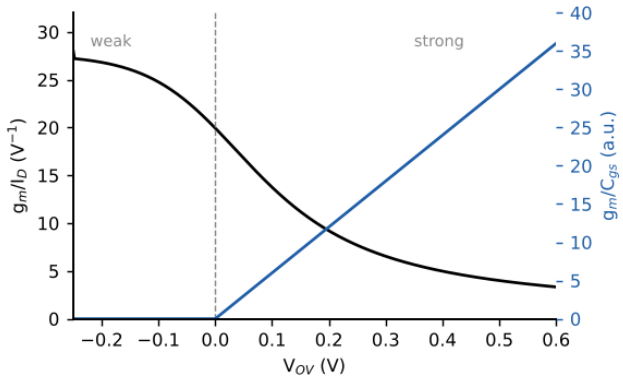
A. 50 GHz
B. 100 GHz
C. 280 GHz
D. 500 GHz

Answer: B. $f_{max} = f \cdot \sqrt{G_p} = 20\ \text{GHz} \times \sqrt{25} = 100\ \text{GHz}$; one measurement at any frequency below f_{max} suffices. (Lecture example: 20 dB at 10 GHz gives 100 GHz.) $\omega_{max} = \tfrac{1}{2}\sqrt{(\omega_T/(r_{gate}C_{gd}))}$ is where this gain reaches unity.

Question 48

L4 Performance Measures of a MOS · gm/ID · Medium

Which statement about g_m/I_D is correct?



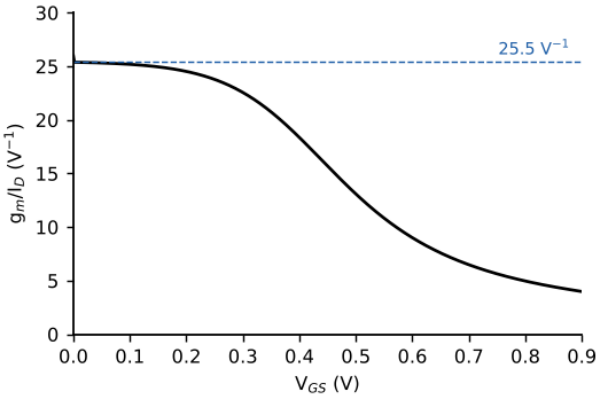
A. $1/V_{OV}$ in strong inversion; it grows without bound in weak inversion
B. Largest in strong inversion, where g_m is largest
C. Independent of bias, since g_m and I_D scale together
D. $2/V_{OV}$ in strong inversion; it saturates at $1/(nU_T)$ in weak inversion, its maximum

Answer: D. Subthreshold $I_D \propto \exp(V_{GS}/nU_T)$ gives $g_m = I_D/(nU_T)$, about 25–30 V^{-1} for $n \approx 1.3$–1.5, while the square law gives $2/V_{OV}$, which falls as V_{OV} rises. V_{OV} is the knob that trades power efficiency (g_m/I_D) for speed (g_m/C_{gs}).

Question 49

L4 Performance Measures of a MOS · Subthreshold swing · Medium

A g_m/I_D simulation in weak inversion gives n ≈ 1.52 at 300 K (U_T ≈ 25.85 mV). The subthreshold swing is approximately:



A. 39 mV/dec
B. 60 mV/dec
C. 90 mV/dec
D. 120 mV/dec

Answer: C. SS = n·U_T·ln10 = 1.52 × 25.85 mV × 2.303 ≈ 90 mV/dec, matching the direct measurement 1000/deriv(log10(I_D)) in the lecture. n = 1 + C_{dep}/C_{ox} measures how much of the channel the body, rather than the gate, controls.

Question 50

L4 Performance Measures of a MOS · Current density · Medium

At fixed I_D and L, reducing the width W (raising the current density I_D/W):

A. Raises g_m/I_D and g_m, and lowers f_T
B. Leaves V_{OV} and g_m unchanged, since I_D is fixed
C. Raises g_m and f_T, because $g_m \propto 1/W$
D. Lowers g_m/I_D and g_m, reduces C_{gg}, and generally raises f_T

Answer: D. $V_{OV} \approx \sqrt{(2L(I_D/W)/(\mu C_{ox}))}$ rises, so $g_m = 2I_D/V_{OV}$ falls, while $C_{gg} \propto W$ falls faster: $f_T = g_m/(2\pi C_{gg})$ rises. Weak inversion is efficient but slow, strong inversion fast but inefficient; moderate inversion is the sweet spot.

Question 51

L4 Performance Measures of a MOS · gm/ID sizing · Medium

The practical g_m/I_D sizing sequence is:

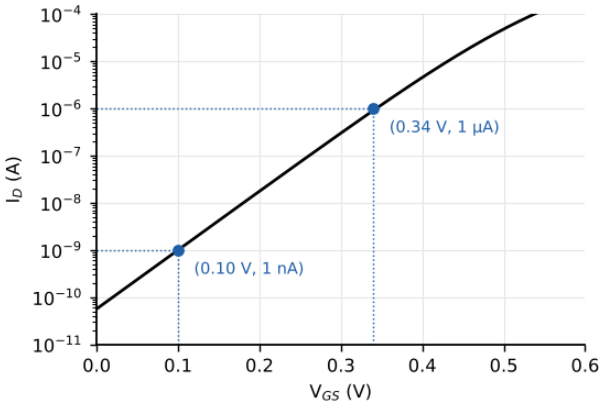
A. Required g_m → choose g_m/I_D and L → $I_D = g_m/(g_m/I_D)$ → $W = I_D/(I_D/W)$, reading I_D/W from the device table
B. Choose W and L, compute I_D from the square law, then read g_m from the table
C. Choose V_{OV}, size W from the square law and adjust it in simulation until the gain is met
D. Choose the minimum L and W, then adjust V_{GS} until the gain and bandwidth are met

Answer: A. Current density is what turns the method into an actual transistor size: g_m/I_D says how efficiently the bias produces transconductance, I_D/W says how heavily the device is biased per unit width. The table (I_D/W, g_m/g_{ds}, C_{gg}) replaces the square-law equations, which are accurate only in a narrow strong-inversion range.

Question 52

L4 Performance Measures of a MOS · I–V curves · Medium

The figure shows $\log(I_D)$ versus V_{GS} of a MOSFET in weak inversion: I_D = 1 nA at V_{GS} = 0.10 V and 1 µA at V_{GS} = 0.34 V. The subthreshold swing and the non-ideality factor n (300 K) are approximately:



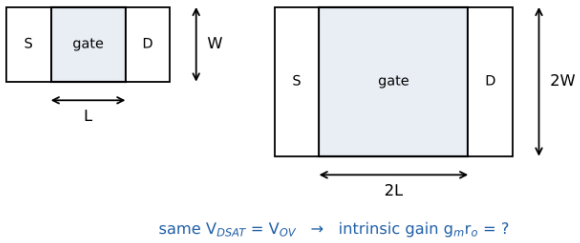
A. 80 mV/dec, n ≈ 1.35
B. 60 mV/dec, n ≈ 1.0
C. 120 mV/dec, n ≈ 2.0
D. 240 mV/dec, n ≈ 4.0

Answer: A. Three decades of current over 0.24 V give SS = 240 mV/3 dec = 80 mV/dec, and $n = SS/(U_T \cdot \ln 10)$ = 80/59.5 ≈ 1.35.

Question 53

L4 Performance Measures of a MOS · Intrinsic gain · Medium

Both the width and the length of a MOSFET are doubled (figure) while $V_{DSAT} = V_{OV}$ is kept constant. Its intrinsic gain $g_m r_o$:



A. Is unchanged, because the aspect ratio W/L and the overdrive are the same
B. Roughly halves, because the larger device carries more current and r_o falls
C. Quadruples, because both g_m and r_o double
D. Roughly doubles: W/L and therefore I_D and g_m are unchanged, but $\lambda \propto 1/L$ halves, so $r_o = 1/(\lambda I_D)$ doubles

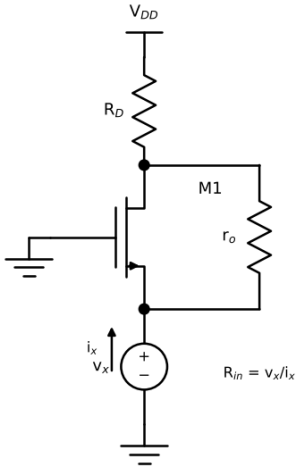
Answer: D. With W/L fixed, $I_D = \tfrac{1}{2}\mu C_{ox}(W/L)V_{OV}^2$ and $g_m = \mu C_{ox}(W/L)V_{OV}$ do not change. Channel-length modulation weakens with length, $\lambda \propto 1/L$, so the Early voltage doubles and $g_m r_o \approx 2/(\lambda V_{OV})$ doubles. The price is speed: $C_{gs} \propto WL$ quadruples, so $\omega_T = (3/2)\mu V_{OV}/L^2$ falls by 4.

Hard (Questions 54–63)

Question 54

L1 Prerequisites Recap · Common-gate stage · Hard

For a common-gate stage with drain load R_D and finite r_o (gate at AC ground, body effect neglected), the exact input resistance looking into the source is:



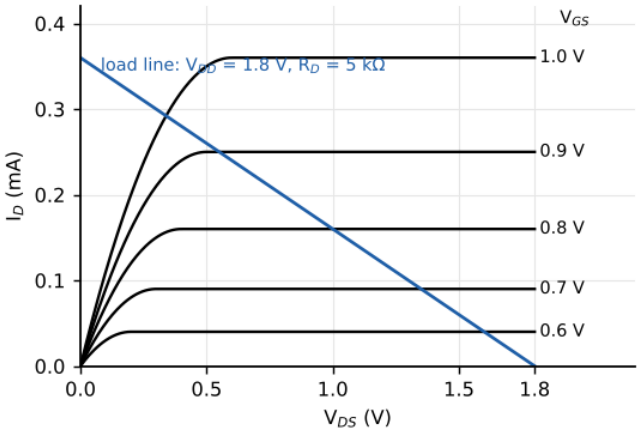
A. $1/g_m$
B. $(1 + g_m r_o)/(r_o + R_D)$
C. $r_o \parallel R_D$
D. $R_D/(g_m r_o)$
E. $(r_o + R_D)/(1 + g_m r_o)$

Answer: E. Drive the source with v_x: $v_{gs} = -v_x$ and $i_x = g_m v_x - (v_d - v_x)/r_o$ with $v_d = i_x R_D$, giving $R_{in} = (r_o + R_D)/(1 + g_m r_o)$. For $r_o \gg R_D$ this is about $1/g_m$; a large load resistance raises it toward $1/g_m + R_D/(g_m r_o)$.

Question 55

L1 Prerequisites Recap · I–V curves · Hard

A common-source stage with $R_D = 5\ k\Omega$ and $V_{DD} = 1.8\ V$ uses the NMOS whose output characteristics are shown ($V_{TH} = 0.4\ V$, $K = 2\ mA/V^2$, $\lambda = 0$); the load line is drawn. With the gate biased at $V_{GS} = 0.8\ V$, the quiescent V_{DS} and the small-signal voltage gain are approximately:



A. $V_{DS} \approx 1.0\ V$, $A_v \approx -8$

B. $V_{DS} \approx 1.0\ V$, $A_v \approx -4$

C. $V_{DS} \approx 0.8\ V$, $A_v \approx -4$

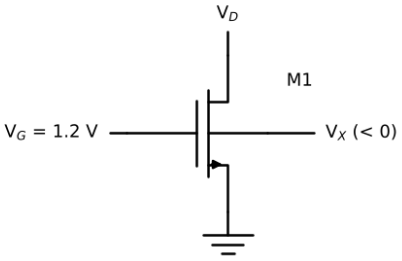
D. $V_{DS} \approx 1.4\ V$, $A_v \approx -2$

Answer: B. The $V_{GS} = 0.8\ V$ curve meets the load line at $I_D = 0.16\ mA$, so $V_{DS} = 1.8 - 0.16\ mA \times 5\ k\Omega = 1.0\ V$, in saturation ($1.0\ V > V_{OV} = 0.4\ V$). $g_m = K \cdot V_{OV} = 0.8\ mS$ and $A_v = -g_m R_D = -4$.

Question 56

L3 Transistor Impairments · Body effect · Hard

An NMOS has $V_{TH0} = 0.3\ V$, $\gamma = 0.4\ V^{½}$ and $2\Phi_F = 0.7\ V$. Its gate is at 1.2 V and its source at ground, while the bulk is driven to a negative voltage V_X. The device turns off once V_{TH} reaches 1.2 V, which happens for V_X below approximately:



A. −0.7 V

B. −1.2 V

C. −3.1 V

D. −8.8 V

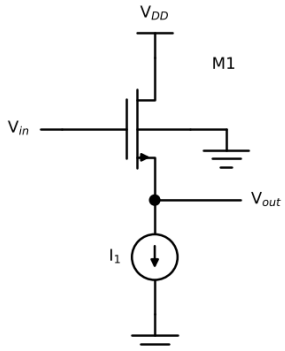
E. −12 V

Answer: D. Solve $1.2 = 0.3 + 0.4(\sqrt{(0.7 - V_X)} - \sqrt{0.7})$: $\sqrt{(0.7 - V_X)} = 2.25 + 0.837 = 3.087$, so $0.7 - V_X = 9.53$ and $V_X \approx -8.8\ V$. The body is a weak gate: almost 9 V of reverse body bias is needed to raise V_{TH} by 0.9 V (at $V_{SB} = 1\ V$ it is only 0.49 V).

Question 57

L3 Transistor Impairments · Body effect · Hard

A source follower is biased by an ideal current source I_1, with the NMOS bulk at ground. As V_{in} rises, what does the body effect do?



A. Nothing: V_{TH} is fixed by the process, so V_{out} tracks V_{in} with a constant V_{GS} shift

B. V_{SB} falls, so V_{TH} falls and V_{out} rises more than V_{in} (gain above 1)

C. V_{SB} rises, so V_{TH} rises and $V_{GS} = V_{in} - V_{out}$ must grow to keep $I_D = I_1$: the gain falls below 1

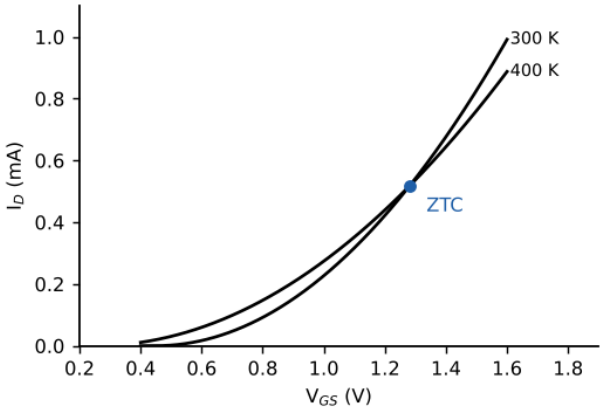
D. It matters only in triode, since in saturation the drain current does not depend on V_{TH}

Answer: C. Body effect appears whenever the source moves relative to the bulk. The level shift V_{GS} is no longer constant; in small-signal terms g_{mb} loads the source node and the gain becomes $g_m/(g_m + g_{mb}) = 1/(1 + \eta)$ for an ideal current-source load (r_o neglected).

Question 58

L3 Transistor Impairments · Temperature dependence · Hard

An NMOS with V_{TH} = 0.4 V is biased at V_{GS} = 1.0 V, i.e. V_{OV} = 0.6 V, below the zero-temperature-coefficient overdrive $V_{OV,ZTC}$ ≈ 0.8 V (figure), so its current rises with temperature. Keeping the same drain current, which change gives I_D a negative temperature coefficient?



A. Double W/L and re-bias: the same current then needs a smaller V_{OV}, which moves the device below $V_{OV,ZTC}$

B. Halve W/L and re-bias: the same current then needs V_{OV} ≈ 0.85 V, above $V_{OV,ZTC}$, where the mobility term dominates

C. Double both W and L: W/L and V_{OV} are unchanged, so the temperature coefficient becomes negative through the longer channel

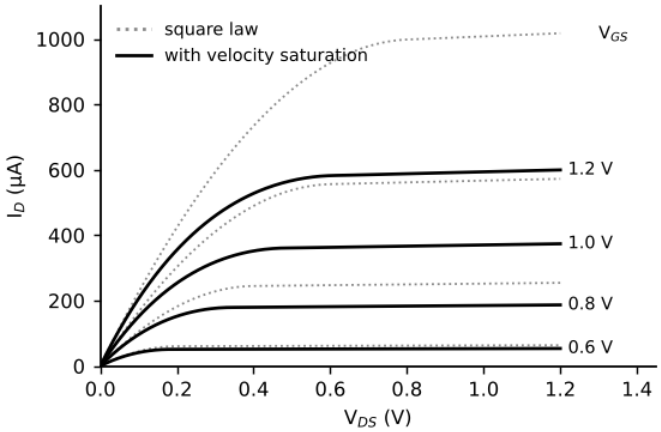
D. Nothing can be done by sizing, because $V_{OV,ZTC}$ itself scales with W/L

Answer: B. $V_{OV,ZTC} = 2|dV_T/dT| \cdot T/(3/2)$ ≈ 0.8 V depends only on the mobility exponent and dV_T/dT, not on geometry. At fixed I_D, $V_{OV} = \sqrt{(2I_D/K)} \propto 1/\sqrt{(W/L)}$: halving W/L raises V_{OV} from 0.6 V to $0.6 \cdot \sqrt{2}$ ≈ 0.85 V, above ZTC, so I_D falls with T. A larger W/L does the opposite, and scaling W and L together changes neither W/L nor V_{OV}.

Question 59

L3 Transistor Impairments · Velocity saturation · Hard

With velocity saturation, $I_D = K[V_{GT}V_{DS} - \tfrac{1}{2}V_{DS}^2] / (1 + V_{DS}/(E_cL))$, with $V_{GT} = V_{GS} - V_{TH}$ and $K = \mu C_{ox}W/L$. Compared with the square law, the I_D–V_{DS} curves show:



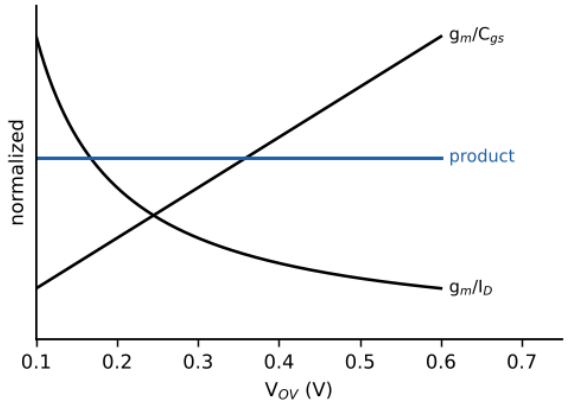
A. Higher saturation current at high overdrive, with V_{DSAT} moving to a higher voltage
B. The largest penalty at low overdrive, where the lateral field is highest, and none at high overdrive
C. Lower saturation current at high overdrive (960 → 600 μA at V_{GS} = 1.2 V) and V_{DSAT} moving left (0.9 → 0.56 V)
D. I_D independent of L, since the denominator cancels the 1/L in K

Answer: C. The penalty appears only where the lateral field is large (high V_{OV}, short L); at V_{GS} = 0.4 V the two curves coincide, so low-overdrive operation is almost unaffected. Saturation begins before pinch-off, and g_m flattens toward $W{\cdot}C_{ox}{\cdot}v_{sat}$.

Question 60

L4 Performance Measures of a MOS · Figures of merit · Hard

In strong inversion, at fixed V_{OV}, the channel length is halved. How do g_m/I_D, g_m/C_{gs} and their product change?



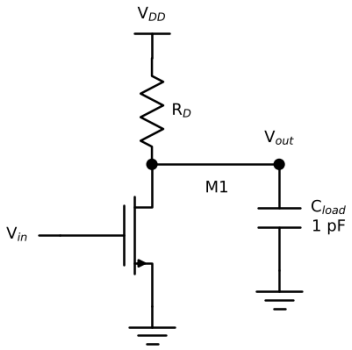
A. g_m/I_D unchanged, g_m/C_{gs} ×2, product ×2
B. g_m/I_D ×2, g_m/C_{gs} ×2, product ×4
C. All three unchanged, since only V_{OV} sets the trade-off
D. g_m/I_D ×½, g_m/C_{gs} ×4, product ×2
E. g_m/I_D unchanged, g_m/C_{gs} ×4, product ×4

Answer: E. $g_m/I_D = 2/V_{OV}$ contains no L; $g_m/C_{gs} = (3/2)\mu V_{OV}/L^2$ quadruples; the product $3\mu/L^2$ quadruples. V_{OV} only trades one figure of merit for the other, whereas a shorter L improves both.

Question 61

L4 Performance Measures of a MOS · gm/ID design · Hard

In the lecture's common-source design example (C_{load} = 1 pF, $|A_v| \geq 30$, BW = 2.5 MHz, L = 0.5 µm), the candidates g_m/I_D = 5, 12 and 20 V^{-1} were compared. $g_m/I_D \approx 12$ was chosen because:



A. 5 fails the gain spec and burns the most current; 20 makes W and C_{gg} large with f_T only about 8× above the GBW; 12 meets the gain with margin at moderate current
B. It gives the highest f_T (about 7 GHz) and the smallest device, so it is the fastest of the three candidates
C. It gives the lowest current (about 31 µA) and the smallest C_{gg}, so it is the most power-efficient of the three
D. The device table at L = 0.5 µm was characterized only at g_m/I_D = 12, so no other value could be used

Answer: A. Moderate inversion balances gain, speed and power. The table gives, for g_m/I_D = 5, 12 and 20: $I_D \approx$ 126, 52 and 31 µA; W ≈ 2.5, 6.5 and 21 µm; $C_{gg} \approx$ 14, 37 and 120 fF; $f_T \approx$ 7, 2.7 and 0.8 GHz; $g_m/g_{ds} \approx$ 25, 40 and 45. At g_m/I_D = 12, f_T is about 27× the GBW; the ratios use GBW = 40 × 2.5 MHz = 100 MHz, the gain actually achieved (33% above the required 30).

Question 62

L4 Performance Measures of a MOS · Current density · Hard

At fixed drain current I_D (square law, saturation), how does f_T scale with the width W?

A. $g_m \propto W$ and $C_{gs} \propto W$, so f_T is independent of W
B. $g_m \propto \sqrt{W}$ while $C_{gs} \propto W$, so $f_T \propto 1/\sqrt{W}$: a narrower device is faster
C. $g_m \propto W$ while $C_{gs} \propto \sqrt{W}$, so $f_T \propto \sqrt{W}$: a wider device is faster
D. $g_m \propto 1/W$ and $C_{gs} \propto W$, so $f_T \propto 1/W^2$

Answer: B. $g_m = \sqrt{(2\mu C_{ox}(W/L)I_D)} \propto \sqrt{W}$ and $C_{gs} = \tfrac{2}{3}WLC_{ox} \propto W$, so $f_T = g_m/(2\pi C_{gs}) \propto \sqrt{W}/W$. Narrowing W at fixed I_D raises the current density and V_{OV}, trading g_m/I_D for speed.

Question 63

L4 Performance Measures of a MOS · Subthreshold swing · Hard

The GPDK090 PMOS model reports a subthreshold swing of 57–59.3 mV/dec for L ≤ 0.2 µm, improving as the channel shortens. The correct interpretation is:

A. An excellent short-channel PMOS, whose electrostatics improve as L shrinks and whose n approaches 1
B. Evidence that holes have a higher mobility than electrons in this process, which lowers n below 1
C. Proof that the NMOS model, which reports 94 mV/dec at L = 0.1 µm, is the one that is wrong
D. A model artifact: with n ≥ 1, SS cannot fall below $U_T \cdot \ln 10 \approx 59$–60 mV/dec at room temperature, and real short channels degrade SS

Answer: D. $n = 1 + C_{dep}/C_{ox}$ cannot be below 1. The NMOS behaves physically: 93.7 mV/dec at L = 0.1 µm, relaxing to about 71 mV/dec once DIBL stops dominating. The 57 mV/dec values and the improvement with shorter L are unphysical, a sign that the PMOS model's weak-inversion region should not be trusted.